

Roll No.

TDM-881

**B. TECH. (CSE/CE & CE-SPL/CST, CST-SPLI &
CST-SPL2/AI&DS) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024**

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Focus on major natural disaster-prone areas in India and discuss the specific risks and challenges faced by these regions. Provide examples of significant natural disasters in India, with a detailed analysis of the Uttarakhand disaster. (CO2)
- (b) Define Disaster, Hazard, Vulnerability, Resilience, and Risks in the context of disaster management. Explain their interrelation. (CO1)
- (c) Discuss the causes, distribution, hazardous effects, and environmental impacts of natural disasters like earthquakes, floods, cyclones, and landslides. (CO6)
2. (a) Define and discuss various man-induced disasters like water logging, groundwater depletion, and release of toxic gases. (CO2)
- (b) Evaluate the differential impacts of disasters on various demographic groups including the elderly, women, children, and people with disabilities. Why do these groups face greater ? (CO5)
- (c) Analyze the impacts of development projects like dams and embankments on local ecosystems and communities. How can these

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projects both mitigate and exacerbate the effects of natural disasters ?

(CO4)

3. (a) Discuss the role of climate change adaptation in reducing disaster risks. What are the key strategies that governments and communities can implement to enhance resilience to climate-related disasters ? (CO6)
- (b) Define sustainable development and discuss its role in disaster mitigation. How can sustainable practices contribute to reducing disaster risks and vulnerabilities ? (CO1)
- (c) Explain the significance of accurate forecasting and warning systems in disaster management. What technologies and methods are most effective for predicting natural disasters ? (CO5)
4. (a) Identify the role of land use zoning in disaster risk reduction. How can urban planning help mitigate the impacts of disasters ? (CO3)
- (b) Classify the effectiveness of Information, Education, and Communication (IEC) campaigns in increasing disaster preparedness. What elements make these campaigns successful ? (CO4)
- (c) What are the key principles of rescue training for search and operation at national and regional levels ? Identify these operations differ between various types of disasters ? (CO3)
5. (a) What are the structure and functional responsibilities of the National Disaster Management Authority (NDMA) ? How does NDMA coordinate with other government and non-government entities ? (CO1)
- (b) Explain the challenges faced by disaster management institutions in India in terms of policy implementation and operational coordination during large-scale disasters. (CO2)
- (c) Apply the key provisions of the Disaster Management Act, 2005. How does this act define the roles and responsibilities of government agencies in disaster management ? (CO3)